

industrial environment owing to their numerous benefits including flexibility, cabling, cost reduction and extending measurements to critical information, amongst others.

In process or manufacturing facilities, the data acquisition system was traditionally limited to a local area or a location in the plant due to the practical difficulty of having communication media running across hostile areas. Now with wireless sensors and I/Os, this is no longer a constraint in having sensor-level data available at any location inside the wireless range.

Powerful new wireless networking technologies have recently become available and are likely to find widespread use in a range of industrial applications. There would be significant applications for Bluetooth and IEEE 802.11b wireless standards in climate and industrial controls, industrial automation, and a variety of mobile systems and equipment.

Wireless technology provides a cost-effective way to monitor plant equipment and production processes, and enables real-time decision-making to optimise production or head-off maintenance issues before these interrupt production. Wireless technology and devices can be used to enable the millions of field devices installed in a process manufacturing facility over the past few years to share process and maintenance information. This information will benefit operations, maintenance and business systems across the enterprise.

However, the rate of adoption for industrial wireless products has been slow due to the challenges and concerns among end users, important ones being reliability, security and safety.

Embedded intelligence and M2M communications. Embedded intelligence is providing manufacturers with improved networking, monitoring and real-time control over plant operations resulting in improved accuracy and increased productivity. It also helps in integrating the control of devices as well as data from devices.

Embedded intelligence facilitates machines to conduct self-diagnostic tests

ments and shorter response times make it essential for any business to invest in tools that would facilitate reliable monitoring of key process and equipment performance parameters for geographically distributed assets.

Today, many companies have a centralised team monitoring the performance of assets spread across different plants that takes decisions to improve outputs and reduce costs. A Web-based application helps ensure availability of relevant information no matter which part of the globe an asset is. This is going to be used extensively in the coming years.

With many of the industrial controllers and communication devices having a Web-enabled server embedded into the card, real-time data is always available in the boardroom. Smart analysis tools and reports hosted on the Web server complementing

the real-time data made available to the management help in quick decision-making and productivity monitoring.

Robots. Use of robotics is one of the major advancements in the field of industrial automation. Robots have taken over many of the operations in manufacturing facilities, increasing the productivity and accuracy as well as reducing the throughput time and operational accidents phenomenally.

Robots are today extensively used for activities that are repetitive or require a great degree of precision. These have also been deployed in environments that are not conducive for human presence. High degree of reliability and consistent quality of output have helped in accelerating the



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to determine their performance efficiency. Microprocessor-based devices lend machines with basic intelligence to compute and control their performance depending on the nature of the job that needs to be executed. Communication across up-stream and down-stream machines helps to ensure that different machines working on the same manufacturing line synchronise their operations thereby eliminating wastage and unnecessary pile-up of inventory.

Web services and applications integration for real-time information. Accurate and timely availability of information using process automation and different communication methods is going to be a key differentiator going forward. Changing customer require-